

Software Engineering

- UE24CS341A

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Problem Statement

Alumni Mentorship & Mock Interview Platform

The Alumni Mentorship & Mock Interview Platform connects students with alumni working in specialized domains. The system provides intelligent mentor matching based on domain expertise, availability-based session booking, and structured post-interview scorecard recording. The main stakeholders are Student Mentee and Alumni Mentor.

Actors

Actor	Description
Student Mentee	Searches for suitable mentors, views availability, books sessions, and views interview feedback.
Alumni Mentor	Provides mentorship/mock interviews, manages availability, and submits interview scorecards.
Platform Administrator	Manages user accounts and platform operations.

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall recommend alumni mentors to students based on domain matching and facilitate one-click session booking.	High	Pass: The system displays at least one eligible alumni mentor whose expertise matches the student's selected domain and allows the student to initiate a booking from the recommendation.	Core functionality for connecting students with suitable alumni mentors.

				Fail: No matching mentor recommendation is displayed when an eligible mentor exists.	
FR-002	Functional	The system shall allow students to view the available dates and time slots of an alumni mentor before booking a session.	High	Pass: When a student selects an alumni mentor, the system displays the mentor's available booking slots. Fail: The system allows booking without displaying the mentor's availability.	Allows students to select a suitable time and prevents scheduling conflicts.
FR-003	Functional	The system shall allow a student to book an available mentorship or mock interview session with an alumni mentor.	High	Pass: When a student selects an available slot and confirms the booking, the session is added to the student's and mentor's calendars and a meeting link is generated. Fail: The booking is not recorded in both calendars or the system permits booking an already occupied slot.	Provides the core scheduling functionality of the platform.
FR-004	Functional	The system shall allow the alumni mentor to record a structured scorecard containing feedback and performance ratings after a mock interview session.	High	Pass: After a completed mock interview, the mentor can submit the scorecard and the system stores the feedback against the corresponding student session. Fail: The mentor cannot submit or save the scorecard after the session.	Provides structured post-interview evaluation for the student.
FR-005	Functional	The system shall allow a student to view the feedback and scorecard associated with a completed mock interview.	Medium	Pass: A student can open a completed session and view its submitted scorecard and feedback. Fail: The system does not display the scorecard after it has been submitted.	Allows students to understand their performance and identify areas for improvement.

Non-Functional Requirements

NFR-001	Nonfunctional	The system shall ensure that all user profile data and feedback scorecards adhere to privacy guidelines and support soft-deletion within 24 hours of a deletion request.	High	Pass: Privacy/security testing confirms compliance with the defined privacy requirements and a deletion request results in soft-deletion within 24 hours. Fail: User data remains active beyond 24 hours after a valid deletion request.	Protects sensitive student, alumni, profile, and feedback information.
NFR-002	Nonfunctional	The system shall display mentor recommendations and available booking slots within 3 seconds under normal operating conditions.	High	Pass: At least 95% of mentor recommendation and availability requests complete within 3 seconds during performance testing. Fail: More than 5% of requests exceed 3 seconds under the defined normal load.	Provides a responsive user experience when searching for mentors and scheduling sessions.

Use Cases

- UC-01: Find / Recommend Mentor
- UC-02: View Mentor Availability
- UC-03: Book Mentorship Session
- UC-04: Conduct Mock Interview
- UC-05: Submit Interview Scorecard
- UC-06: View Interview Feedback
- UC-07: Manage User Accounts

UML Relationships

- UC-03 Book Mentorship Session «include» UC-02 View Mentor Availability
- UC-05 Submit Interview Scorecard «extend» UC-04 Conduct Mock Interview

UML Use-Case Diagram:

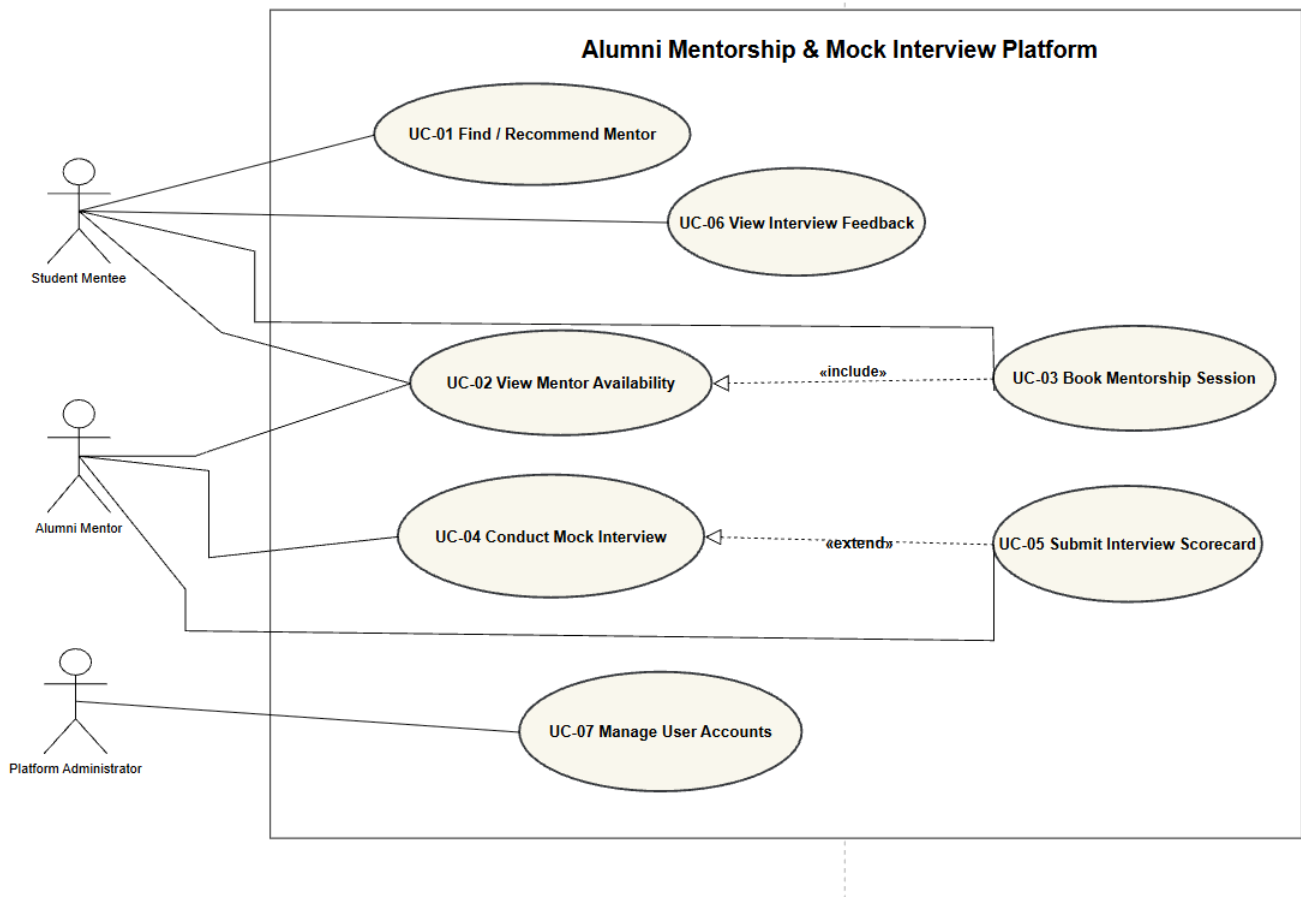


Fig. UML Use-Case Diagram for Alumni Mentorship & Mock Interview Platform

Use-Case Flow Specification

Use Case: Book Mentorship Session

Preconditions:

- Student Mentee is logged into the system.
- Suitable mentor profiles are available.
- Mentor has available time slots.

Main Success Scenario:

1. Student selects a desired domain.
2. System recommends suitable alumni mentors.
3. Student selects a mentor.
4. System displays available time slots.
5. Student selects an available slot.
6. System validates the slot.
7. Student confirms the booking.
8. System creates the session and updates both calendars.

9. System provides the meeting link and booking confirmation.

Alternate Flow — Slot Unavailable:

A1. System detects that the selected slot has already been booked.

A2. System informs the student that the slot is unavailable.

A3. System displays other available slots.

A4. Student selects another slot and continues the booking process.

Postconditions:

- The mentorship/mock interview session is successfully booked.
- The student and mentor calendars are updated.
- A meeting link is generated.